

Python

Let's understand Python.

What is Python programming language?

Python is a **high-level, general-purpose, interpreted** programming language. which is used for:

- web development (server-side),
- software development,
- mathematics,
- system scripting.
- Game Development

And much more

1) High-level

Python is a high-level programming language with readable English like codes that makes it easy to learn. As Python is a high level language you don't need to understand the details of the computer architecture to develop programs efficiently.

Difference between High Level And Low Level Programming Languages

High-level languages **are easy to understand** and **debug**.

Advantages: portable and do not depend on machines.

Low-level languages, are **machine-friendly**, but **difficult to understand**,

Disadvantage: They are not portable. They are machine-dependent and not commonly used for programming.

Examples:

High-level Languages – Python, Visual Basic, Java, C, C++, SQL.....

Low-level Languages – assembly languages and machine code which are Hardware/Processor-specific.

2) General-purpose

Python is a general-purpose language. Python can be used in various domains including:

Web applications

Data science,

machine learning,

AI

Desktop software

Mobile apps

Data applications

Testing

Automation

Python	+	Pandas	=	Data manipulation
Python	+	TensorFlow	=	Deep learning
Python	+	Matplotlib	=	Data visualization
Python	+	Seaborn	=	Advanced charts
Python	+	BeautifulSoup	=	Web scraping
Python	+	Selenium	=	Browser automation
Python	+	FastAPI	=	High-performance APIs
Python	+	SQLAlchemy	=	Database access
Python	+	Flask	=	Lightweight web apps
Python	+	Django	=	Scalable platforms
Python	+	OpenCV	=	Game development

3) Interpreted

To execute the source code written in high level programming language, you need to convert it to the machine language that the computer can understand.

There are two ways to do so

*One by **Compiler***

*Second by **Interpreter***

Difference between Compiler and Interpreter:

Compiler: A compiler translates all lines of code in a program of a high-level programming language into machine code at one go before the program runs.

Interpreter: An interpreter translates code written in a high-level programming language into machine code line-by-line as the code runs.

Python is an interpreted language. To develop a Python program, you write Python code into a file called source code.

And the Python interpreter turns the source code, line by line, once at a time, into the machine code when the Python program executes.

Other languages like Java and C# use a compiler that compiles the whole source code before the program executes.

W H Y P Y T H O N ?	• Python works on different platforms (Windows, Mac, Linux, Raspberry Pi, etc). • Python has a simple syntax similar to the English language. • Python has syntax that allows developers to write programs with fewer lines than some other programming languages. • Python runs on an interpreter system, meaning that code can be executed as soon as it is written. This means that prototyping can be very quick. • Python can be treated in a procedural way, an object-oriented way or a functional way.
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Python Syntax compared to other programming languages

- Python was designed for readability, and has some similarities to the English language with influence from mathematics.
- Python uses new lines to complete a command, as opposed to other programming languages which often use semicolons or parentheses.
- Python relies on indentation, using whitespace, to define scope; such as the scope of loops, functions and classes. Other programming languages often use curly-brackets for this purpose.

For example -

```
1. def func():  
2.     statement 1  
3.     statement 2  
4.     .....  
5.     .....  
6.     statement N
```

In the above example, the statements that are at the same level to the right belong to the function. Generally, we can use four whitespaces to define indentation.

Instead of Semicolon as used in other languages, **Python ends its statements with a NewLine character.**

- **In Python, it is easy to learn and write code.** While in Java, C or C++, it requires more code to perform tasks.

Example

Python Code:

```
print("Hello, World!")
```

Java Code:

1. **public class** HelloWorld {
2. **public static void** main(String[] args) {
3. System.out.println("Hello, World!");
4. }
5. }

In Java, we need to declare classes, method structures and many other things.

While both programs give the same output, we can notice the syntax difference in the print statement.

- **Python is dynamically typed**, meaning we do not need to declare the variable. Whereas most of the other languages are statically typed, meaning we need to declare the variable type.
- **Python is suitable for various domains** such as Data Science, Machine Learning, Web development, and more. Whereas Java is suitable for web development, mobile app development (Android), and more.

Features of Python:

Easy to use and Read - **Python's syntax is clear and easy to read, making it an ideal language for both beginners and experienced programmers. This simplicity can lead to faster development and reduce the chances of errors.**

- **Dynamically Typed** - The data types of variables are determined during run-time. We do not need to specify the data type of a variable during writing codes.
- **High-level** - High-level language means human readable code.
- **Garbage Collected** - Memory allocation and de-allocation are automatically managed. Programmers do not specifically need to manage the memory.
- **Purely Object-Oriented** - It refers to everything as an object, including numbers and strings.
- **Cross-platform Compatibility** - Python can be easily installed on Windows, macOS, and various Linux distributions, allowing developers to create software that runs across different operating systems.
- **Rich Standard Library** - Python comes with several standard libraries that provide ready-to-use modules and functions for various tasks, ranging from **web development** and **data manipulation** to **machine learning** and **networking**.
- **Open Source** - Python is an open-source, cost-free programming language. It is utilized in several sectors and disciplines as a result.

Python has many *web-based assets, open-source projects, and a vibrant community*. Learning the language, working together on projects, and contributing to the Python ecosystem are all made very easy for developers.

- **Python is a case-sensitive language**, which means that uppercase and lowercase letters are treated differently. For example, 'name' and 'Name' are two different variables in Python.
- **In Python, comments can be added using the '#' symbol.** Any text written after the '#' symbol is considered a comment and is ignored by the interpreter. This trick is useful for adding notes to the code or temporarily disabling a code block. It also helps in understanding the code better by some other developers.

Python Installation

Recommended System Requirements

- Processors: Intel Core i5 processor 4300M at 2.60 GHz or 2.59
- 8 GB of DRAM
- Disk space: 2 to 3 GB
- Operating systems: Windows 10, macOS, or Linux

Minimum System Requirements

- Processors: Intel Atom processor **or** Intel Core™ i3 processor
- Disk space: 1 GB
- Operating systems: Windows 7 or later, macOS, or Linux

Python is open source and is available free

Download python setup

Python is open source and is available free on its official website

<https://python.org/downloads/>

Python versions: 3.14.4 (Latest Download from)

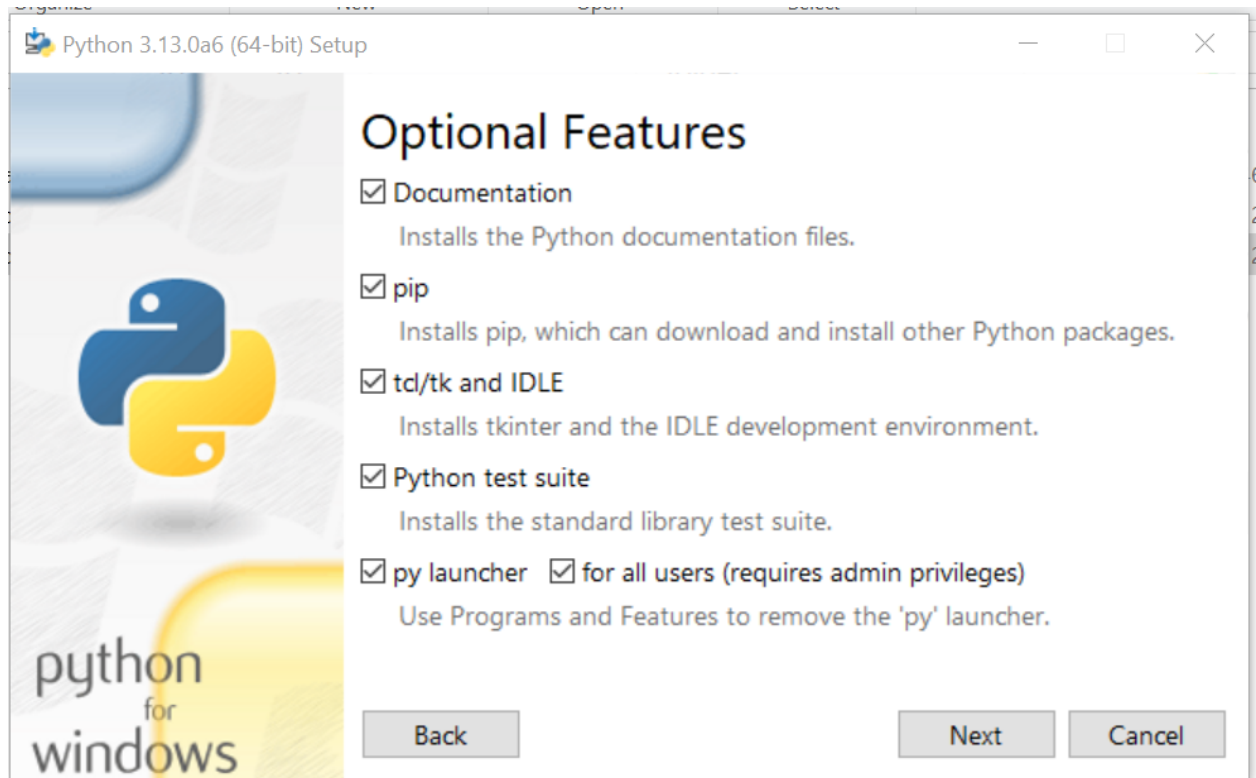
Download python setup and execute the file

Check add python.exe to path

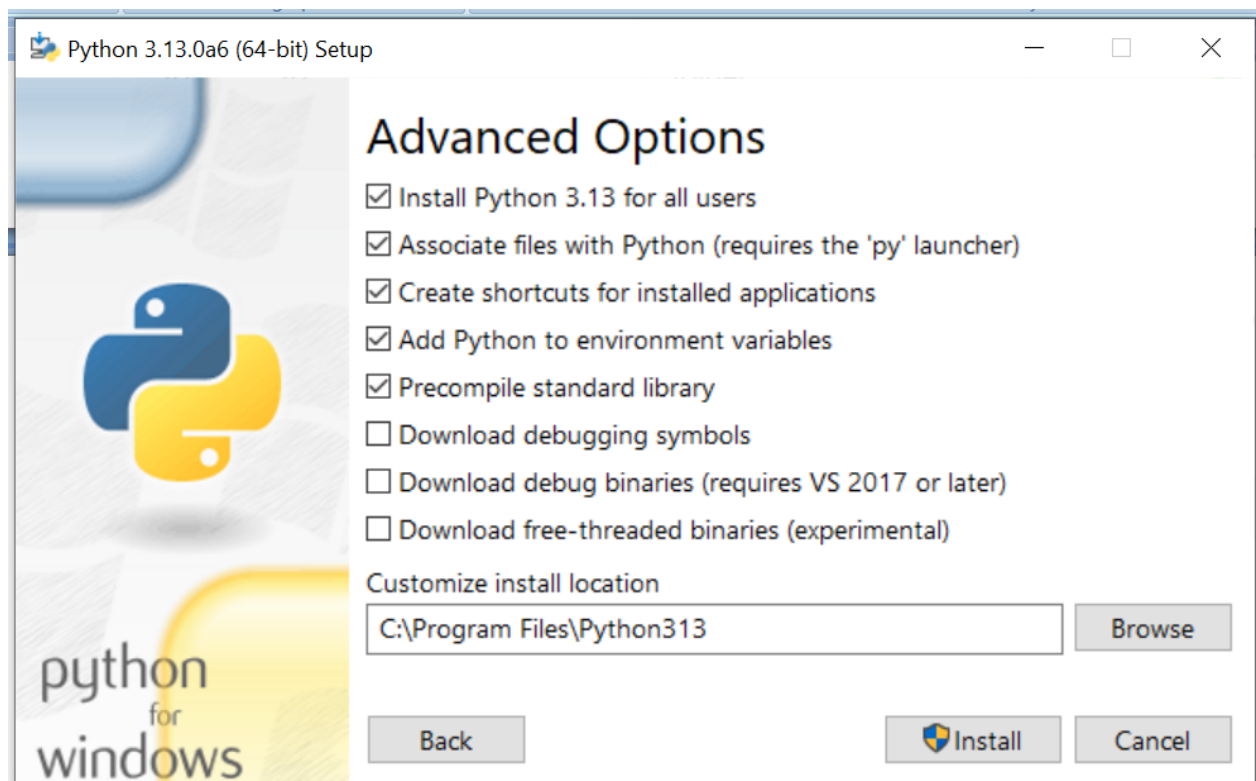


Must check Add python.exe to PATH

Must include pip during setup. Pip helps to install packages like django and other later



Check install for all users.



After successful installation check by typing

python --version

on command prompt in windows

python3 --version

in terminal of mac

Installing IDE

➤ PyCharm

PyCharm **Community version** (Download Link for windows)

<https://www.jetbrains.com/pycharm/download/?section=windows>

Download and install PyCharm

After installation run the program. And select do not import settings

➤ VS Code

Download and install vs code from

<https://code.visualstudio.com/download>

After installation **add extension for python**

Print Function

The `print()` function prints the message on to the screen, or other standard output device.

The message can be a string, a number, a Boolean or any other object, the object will be converted into a string before written to the screen.

Syntax

`print(object(s), sep=separator, end=end, file=file, flush=flush)`

object(s)-- Any object, and as many as you like. Will be converted to string before printed

sep='separator' -- Optional. Specify how to separate the objects, if there is more than one. Default is ' '

end='end' -- Optional. Specify what to print at the end. Default is '\n' (line feed)

file -- Optional. An object with a write method. Default is sys.stdout

flush -- Optional. A Boolean, specifying if the output is flushed (True) or buffered (False). Default is False

Examples

```
print("Hello", "how are you?")
```

```
print("Hello", "how are you?", sep="---")
```

using variables:

```
name = "Python"  
typ = "Programming Language"  
print("Name : ", name)  
print("Type: ", typ)
```

Using Numbers

```
a = 75  
b = 29  
product = a * b  
print(product)
```

input() Function

The `input()` function allows user input.

Syntax

`input(prompt)`

prompt A String, representing a default message before the input

Example

```
name = input('Enter your name:')  
num1 = int(input("Enter a number :"))  
a = float(input("Enter a decimal number :"))
```